

Weather and the people: How different weather features impact health and behavior

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Introduction

The weather is impacting the daily life of everyone. Sometimes it can destroy your thoroughly thought out weekend plans and sometimes it can make for a refreshing change. In other words, it influences our life, probably to a non-negligible extend. But it is difficult to say how much of an impact it really has in the end. That's why we decided to dig through some data to find out. To do that, we collected different topics about health, happiness and the behaviour of people to compare them to different weather features.

Method

Since we covered a lot of different topics, we collected data from a lot of different data sources. The most important one is the OpenMeteo API [4]. We used the historical weather API at OpenMeteo to collect daily weather features in different cities across Germany from 2022 to 2024. Some of those features are mean temperature, precipitation hours, sunshine duration, shortwave radiation and maximum wind speed. We then collected topic specific data from the OpenMeteo Air Quality API [4], the Worldbank Climate Change Knowledge Portal [6], the World Happiness Report [5], the Google Trends API [2] and the Cloudflare Radar API [1], to compare it to our weather data.

Overview

To give you an overview about the climate situation worldwide and the differences between states in Germany, we made the following maps showing different weather features.

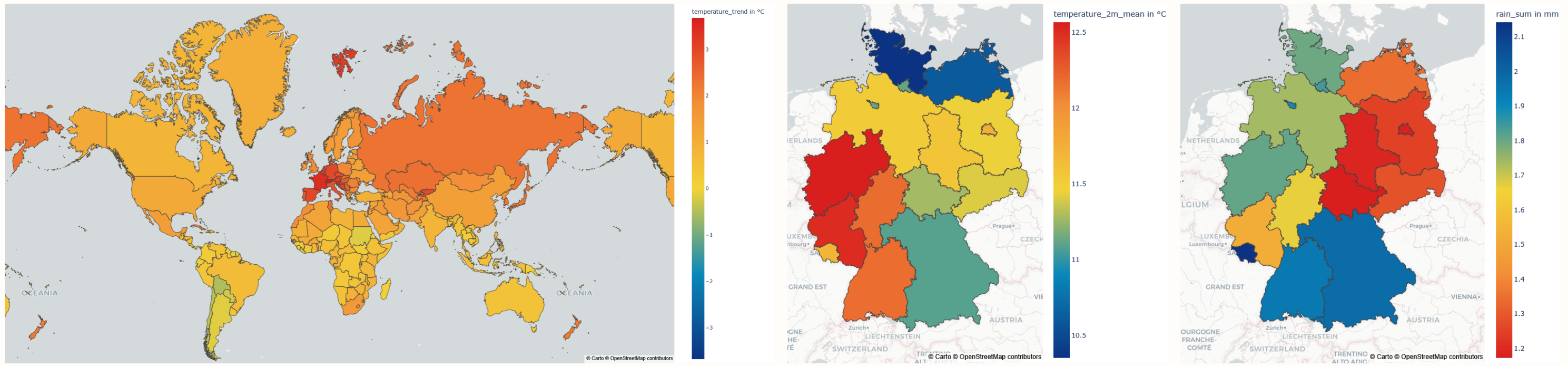


Figure 1: Temperature trend from 1901 to 2022

Figure 2: Average daily temperature from 2022 to 2024

Figure 3: Average daily rain from 2022 to 2024 per state

Effect on consumer behavior

We used Google Trends to compare different keywords in different regions of Germany to the respective weather. The most interesting correlation that we found is between precipitation hours and people searching for "kino", as is shown in the following graphs:

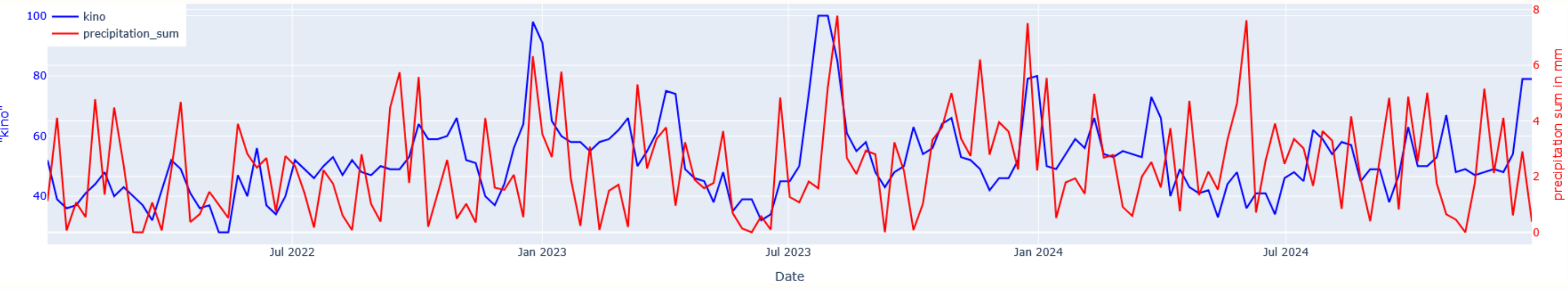


Figure 4: Average precipitation sum per state and "kino" percentage of maximum in google trends in Germany from 2022 to 2024

In fact, we found a weak positive correlation of 22% and a p-value of 0.0047, making the correlation statistically significant.

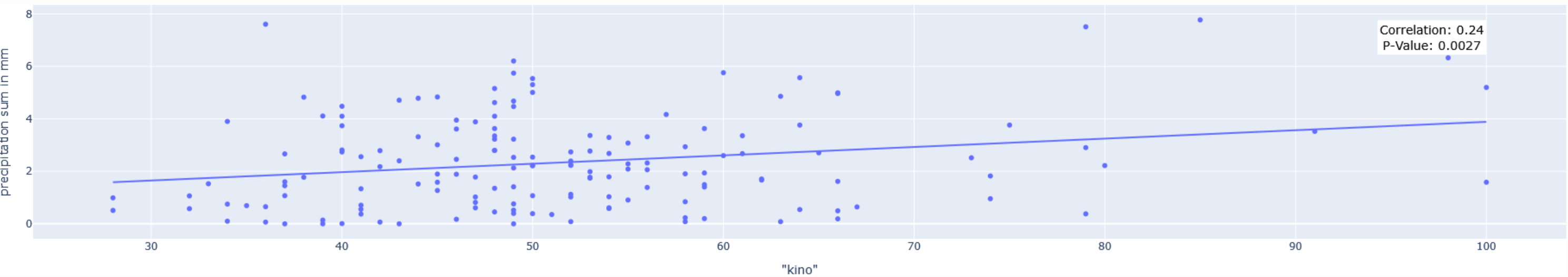


Figure 5: Correlation between average precipitation sum per state and "kino" percentage of maximum in google trends in Germany from 2022 to 2024

Wellbeing

Human well-being is closely influenced by environmental factors. We've looked at these factors, specifically temperature and shortwave radiation, to understand their impact.

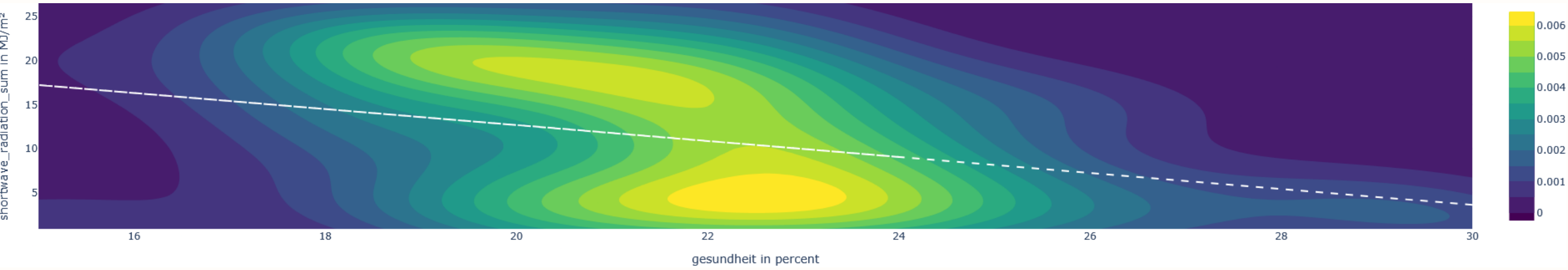


Figure 6: Average daily short wave radiation against "Gesundheit"

The provided graph illustrates the relationship between shortwave radiation and health awareness. The negative correlation, as indicated by the downward trend of the dashed regression line, suggests that increased shortwave radiation exposure is associated with lower health awareness levels. We were able to measure the same trends for the average temperature as well as sunshine duration of the respective states.

Correlation with air quality

Air quality has a major impact on overall life quality. Kong, Liang, and Liu [3] present strong evidence that not only particulate matter but ozone is a major health hazard.

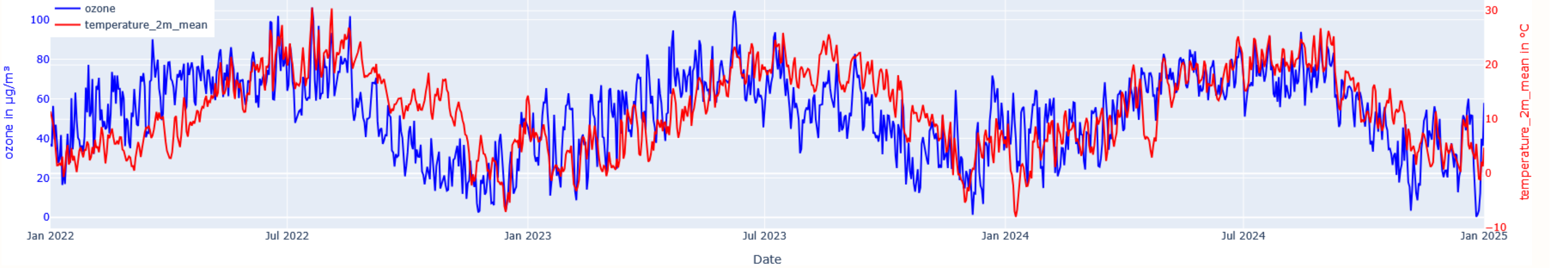


Figure 7: Ozone at 10 m elevation and mean temperature in Berlin from 2022 to 2024

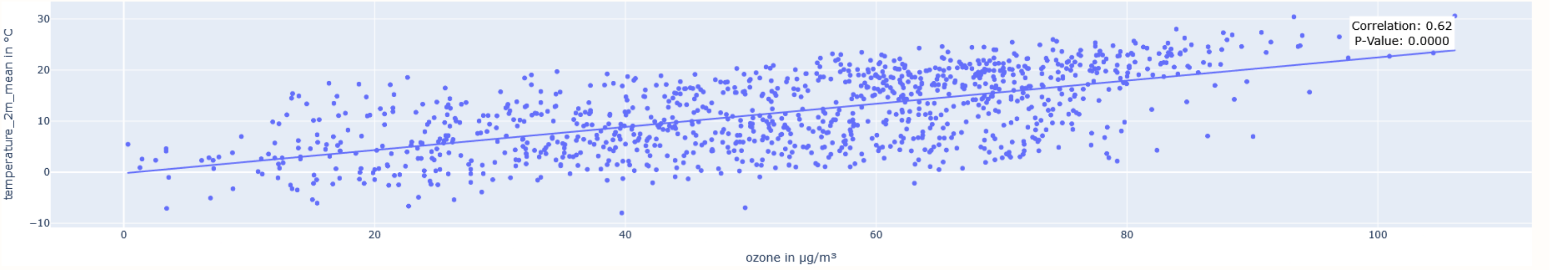


Figure 8: Correlation between ozone level 10 m above surface level and mean temperature in Berlin from 2022 to 2024

As seen in figures 7 and 8 we see a positive correlation between the mean temperature and the amount of ozone 10 m above surface level in Berlin.

Temperature vs. Happiness

Since most of our research so far has been about Germany alone, we thought it might be interesting to also compare different countries. To that end, we used the world happiness score and put it in relation to the average annual temperature of each country.

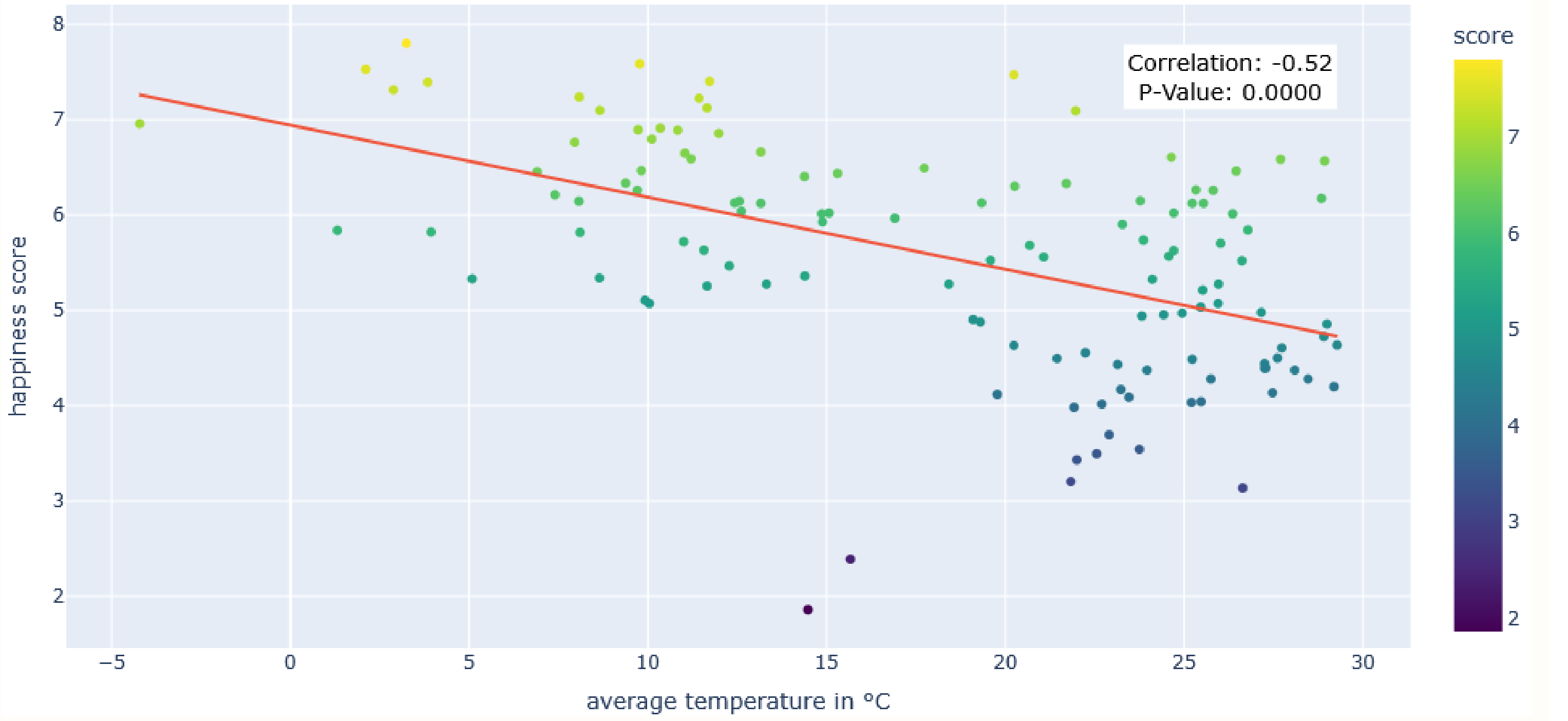


Figure 9: Correlation of mean temperature and happiness score for 2022

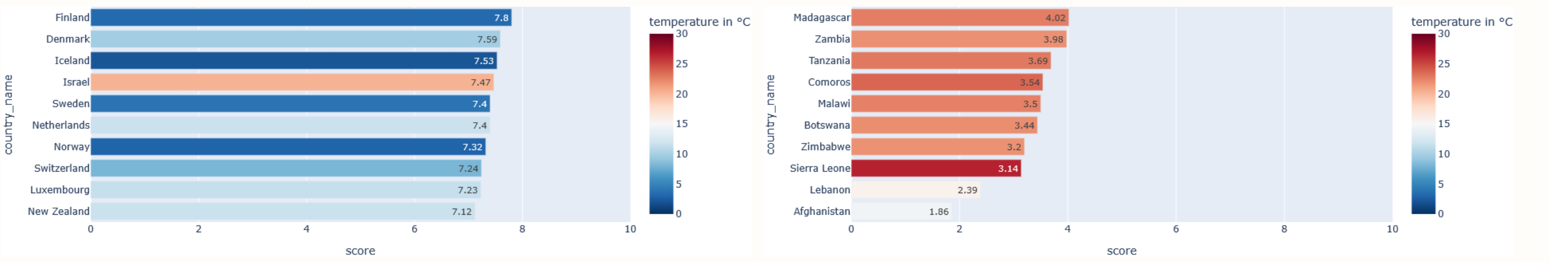


Figure 10: Top 10 Happiest countries in 2022

Figure 11: Top 10 Unhappiest countries in 2022

Notably, we found a negative, statistically significant correlation of 52%, meaning that higher average annual temperature seems to lead to a lower happiness score.

References

[1] Cloudflare. *Cloudflare Radar API*. 2025. URL: <https://radar.cloudflare.com/traffic/de>.

[2] Google. *Google Trends*. 2025. URL: <https://trends.google.com/trends/>.

[3] Dongmin Kong, Junwei Liang, and Chenhao Liu. "Invisible enemy: The health impact of ozone". In: *China Economic Review* 72 (2022), p. 101760. ISSN: 1043-951X. DOI: <https://doi.org/10.1016/j.chieco.2022.101760>. URL: <https://www.sciencedirect.com/science/article/pii/S1043951X22000189>.

[4] OpenMeteo. *Historical Weather API and Air Quality API*. 2025. URL: <https://open-meteo.com>.

[5] Wellbeing Research Centre at the University of Oxford. *World Happiness Report*. 2025. URL: <https://worldhappiness.report>.

[6] Worldbank. *Climate Change Knowledge Portal*. 2025. URL: <https://climateknowledgeportal.worldbank.org/>.

Additional research questions and interactive plots, featuring a wider range of weather features, are available on our website via the QR code in figure 12. Explore different topics and try different settings to find the most interesting information!



Figure 12: QR Code to Render Website